

Real-Time Anomaly Segmentation for Road Scenes

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Abstract

Real-time anomaly segmentation in open-world scenarios is important for safety-critical applications like autonomous driving. Although the latest semantic segmentation networks do a great job with in-distribution data, they often struggle to spot and identify out-of-distribution (OoD) or unknown objects. In this work, we address this limitation by developing lightweight and deployable models that preserve competitive segmentation accuracy while operating under strict real-time and resource constraints. Building on ERFNet as backbone architecture, we evaluate established OoD detection techniques —Maximum Softmax Probability, Max Logit, and Maximum Entropy— enhanced with temperature scaling for improved confidence calibration. Additionally, we investigate the impact of void-class supervision and introduce an extension based on two recent loss functions: Logit Normalization Loss and Enhanced Isotropy Maximization Loss. Our approach is assessed on standard benchmark datasets like Cityscapes, Fishyscapes, and SMIYC RA/RO-21, showing improved anomaly segmentation without compromising efficiency. (<https://github.com/lisagiacob/AML-AnomalySegmentation>)

1. Introduction

With the development of intelligent vehicles (IV) and autonomous driving, it is becoming increasingly important to improve the perception of the environment. Although this typically involves the integration of numerous sensors, cameras have gained a prominent role thanks to the remarkable advances in the computer vision field. Images are a rich multi-dimensional signal that is cheap to capture, but that requires complex algorithms to process.

Traditional vision-based approaches were originally designed to detect traffic elements independently, such as road pavement, pedestrians, cars, traffic signs, or traffic lights (figure 1).

Recent advances in deep learning has made it possible to unify all of these perception tasks into a single one: semantic segmentation [7]. In particular, deep convolutional neural networks (CNNs) excel at this task, as they can be

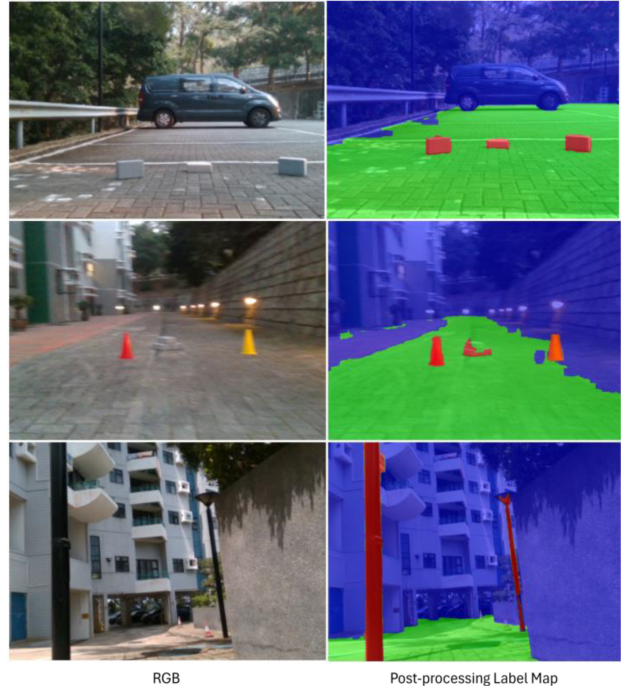


Figure 1. Example of road anomaly detection (image from [5])

trained end-to-end to classify multiple object categories in an image at pixel level [7].

While high-quality semantic segmentation has advanced considerably, research on achieving real-time performance without compromising accuracy has lagged behind [11]. Both low-level details and high-level semantics are crucial for semantic segmentation; nevertheless, many real-time approaches accelerate inference at the cost of discarding spatial detail, which leads to a notable accuracy drop [9].

Another major limitation of state-of-the-art semantic or instance segmentation networks is that they are usually trained on a closed set of semantic classes. As a result, they are ill-equipped to handle previously unseen or out-of-distribution (OoD) objects. Detecting and localizing such unknown objects is, however, essential for safety-critical applications such as autonomous driving, particularly when anomalies appear on the road ahead [1].

In this paper, we address the problem of real-time

anomaly segmentation in open-world contexts, with a particular focus on feasibility on devices with limited resources, such as smart cameras with constrained computing capacity. Our objective is to design and evaluate anomaly segmentation models that are at the same time light, efficient and usable in real-time while still maintaining competitive performance. This is particularly important in edge applications, where computing and memory resources are severely limited.

Starting from existing lightweight architectures such as ERFNet, we conducted a series of experiments with different anomaly detection methods, including Maximum Softmax Probability (MSP), MaxLogit and Max Entropy. We also applied temperature scaling, to improve the calibration of model confidence score with the aim of increasing the accuracy in the identification of anomalies. Another direction we have explored is the use of the void class as an anomaly indicator, using lightweight networks such as ENet and BiSeNet. Finally, as an extension of the project, we analyzed the effect of OoD-specific loss functions, focusing in particular on Logit Normalization Loss and Enhanced Isotropy Maximization Loss, to assess whether anomaly segmentation can be improved without increasing model complexity.

Our main contributions are the following:

- We compared and analyzed different lightweight architectures for anomaly segmentation, evaluating their effectiveness under computational constraint and memory constraints.
- We evaluated different anomaly detection strategies, including calibration techniques such as temperature scaling, assessing their strengths and weaknesses.
- We implemented a classifier based on the void class, testing the ability of light networks to detect objects absent from the training.
- We assessed the impact of OoD-specific loss functions, observing improvements in the model capacity without increasing the number of parameters.
- We reported quantitative results on standard benchmarks for anomaly segmentation (SMIYC RA-21, RO-21, Fishyscapes), comparing methods with metrics such as mIoU, AuPRC and FPR95

2. Related work

Among the various semantic segmentation models, several lightweight architectures have been proposed to balance accuracy and computational efficiency, particularly for real-time deployment systems on edge devices. One of the earliest and most influential is ENet [6], which was specifically designed for real-time tasks. As shown in Figure 2,

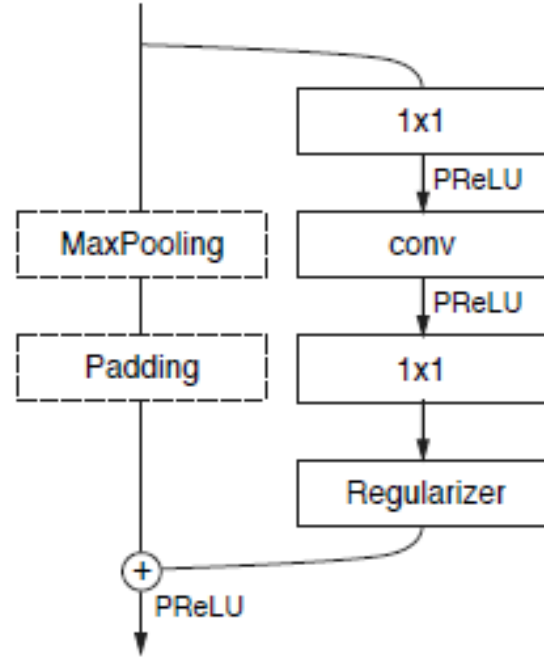


Figure 2. ENet architecture (image from [6])

it's an encoder-decoder structure divided into several stages, adopting a view of ResNets that describes them as having a single main branch and extensions with convolutional filters that separate from it, and then merge back with an element-wise addition. The model achieves impressive efficiency—up to 18× faster, 75× less FLOPs, and 79× less parameters than comparable networks—while maintaining competitive accuracy. This efficiency, however, comes at the cost of reduced precision compared to heavier architectures. [6].

To address these limitations, ERFNet [7] introduced a novel residual layer with factorized convolutions. The core of this architecture (figure 3) is a novel layer that uses residual connections and factorized convolutions in order to remain efficient while retaining remarkable accuracy. Experiments demonstrate that the resulting architecture provides an excellent trade-off between reliability and speed, which makes it suitable for countless IV applications, such as scene understanding in self-driving vehicles that require both robustness and real-time operation [7].

Another notable architecture is BiSeNet [10], which separate the network in two part (figure 4): a *Spatial Path* with a small stride to preserve the spatial information and generate high-resolution features, and a *Context Path* with a fast downsampling strategy to capture a large receptive field. This bilateral design yields a strong balance between accuracy and efficiency, achieving competitive results on Cityscapes, CamVid, and COCO-Stuff

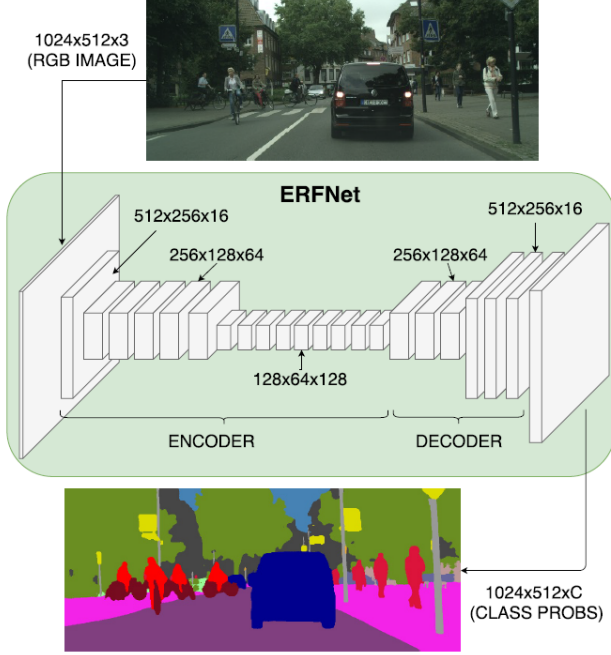


Figure 3. ERFNet architecture (image from [7])

datasets. BiSeNetV2 [9] improves the framework introducing a Guided Aggregation Layer to merge spatial and contextual features effectively.

ICNet [11] employs a cascade strategy processing images at multiple resolutions. It adopts a cascade feature fusion unit and is trained with a label guidance mechanism, achieving high-quality segmentation at real-time speeds, making it another relevant baseline for resource-constrained deployment.

2.1. Anomaly Segmentation and OoD Detection

While lightweight networks enable real-time segmentation, reliability in the presence of unknown objects remains a major challenge, particularly for autonomous driving [1]. This has motivated research in anomaly segmentation, where the objective is not only to detect unknown regions but also to provide pixel-wise masks of anomalous areas.

Several OoD detection strategies have been investigated. The Maximum Softmax Probability (MSP) baseline estimates confidence using the highest softmax score. More recent approaches propose the MaxLogit method, which directly uses the maximum logit value and has shown clear improvements over MSP, and Maximum Entropy, which measures the uncertainty of predictions through the entropy of the softmax distribution. Despite their effectiveness, these methods often suffer from overconfidence on OoD samples. To mitigate this, temperature scaling [2] is widely adopted as a post-processing calibration technique, adjust-

ing the softmax distribution to yield more reliable confidence estimates and improve separation between in- and out-of-distribution data.

2.2. Evaluation Metrics

To evaluate segmentation performance, we use the mean Intersection over Union (mIoU) metric, defined as

$$mIoU = \frac{1}{N} \sum_{c=1}^N IoU \quad (1)$$

where the per-class Intersection over Union is given by

$$IoU = \frac{TP}{TP + FP + FN} \quad (2)$$

The mIoU provides a robust estimate of segmentation quality across classes, independent of class imbalance, computing the mean of the overlap between the predicted regions of the model and the real region (ground truth) over all classes.

For OoD detection, complementary metrics such as AuPRC and FPR95 are also employed, focusing on the ability to discriminate anomalies from in-distribution pixels.

2.3. Datasets

The baseline model, ERFNet, was trained on the **Cityscapes** dataset. Cityscapes consists in high-resolution urban images from 50 different cities, with fine pixel-level annotations for 30 semantically relevant classes. It's designed to support the semantic understanding of urban scenes and represents a standard for training semantic segmentation models in urban environments.

To evaluate our model's ability to detect out-of-distribution (OoD) objects, we used several datasets specifically designed to test performance in autonomous driving scenarios. Among these, Fishyscapes evaluates pixel-level uncertainty estimation in the presence of anomalous objects in urban driving scenarios. In our experiments, we considered two of its subsets: **Lost & Found (FS L&F)** contains urban images with unexpected or rare objects on the road, and **Static (FS Static)**, composed of synthetic images generated from Cityscapes with foreign objects inserted into the scenes.

In addition, we used the datasets introduced by the **SegmentMelfYouCan** benchmark. Specifically, **RoadAnomaly21** provides diverse real-world images featuring anomalous events in road scenarios, testing the model's robustness against unforeseen hazards, while **RoadObstacle21** focuses on obstacles located on the drivable area, allowing a more safety-oriented evaluation of the model's capacity to detect and localize unexpected obstacles.

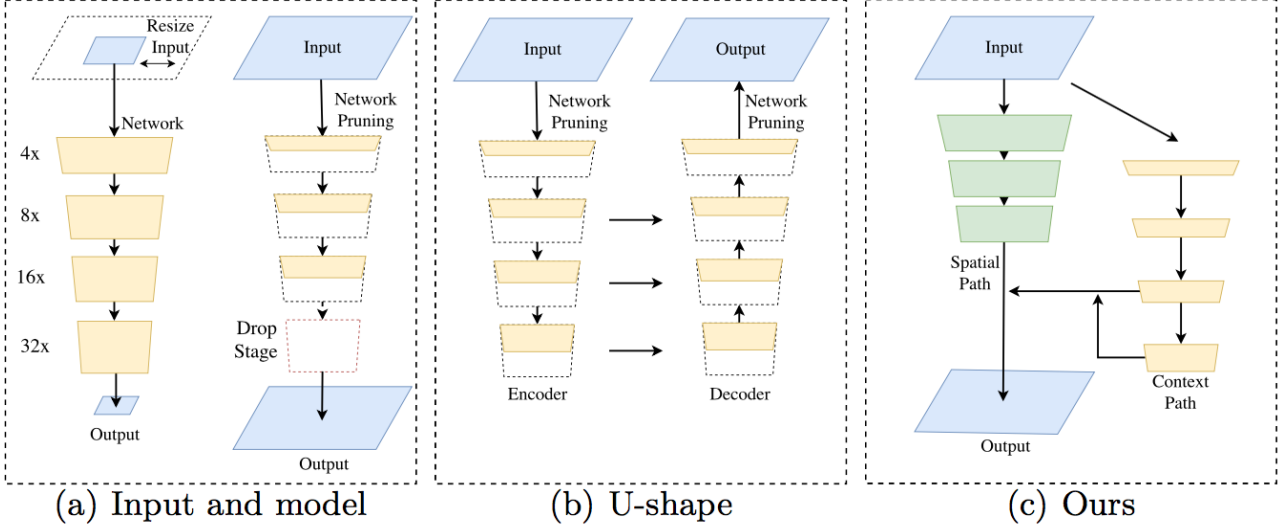


Figure 4. BiSeNet architecture (image from [10])

3. Methodology

In this section, we present the methodology adopted for real-time anomaly segmentation using lightweight architectures. We started from ERFNet and compared different lightweight backbones, evaluate several anomaly detection strategies (including calibration), and finally explore loss-function modifications to enhance OoD performance under deployment constraints.

3.1. Baseline setup

Starting from a pre-trained ERFNet model provided to us by the official repository The network has been trained on Cityscapes, a recent dataset of urban scenes that has been widely adopted in semantic segmentation benchmarks due to its highly varied set of scenarios and challenging set of 19 labeled classes. [7].

Using a pre-trained ERFNet model on Cityscapes allows to immediately have a robust semantic segmentation backbone, avoiding the training phase from scratch, which would be computationally burdensome. In this way, it is possible to focus the analysis on anomaly inference, directly evaluating techniques without introducing variables related to training quality.

In particular, we focus on three main techniques:

- MSP (Maximum Softmax Probability) exploits the maximum softmax probability to indicate confidence. Low values highlight anomalies.
- Max Entropy uses softmax probability on all classes and calculates their entropy. High entropy indicates that the model is less secure .

- Max-Logit exploits the maximum value of pre-softmax logits, avoiding the distortion introduced by normalization. Low logits correspond to low confidence and, therefore, possible anomalies.

We evaluated the application of this methods on the different models based on two parameters:

- AuPRC: the area of a curve that connects precision and recall. The higher its value, the more the model is capable of detecting anomalies.
- FPR95 expresses the rate of false positives, the lower its value, the lower the number of false positives, thus obtaining lower noise.

MaxLogit emerges from the results, collected in Table 1, as the most consistent method, with improved AuPRC and/or lower FPR95 on key benchmarks (e.g., SMIYC RA-21: AuPRC 38.319 vs. 29.100 MSP and 31.005 Max Entropy). MSP sometimes attains competitive AuPRC but often at the cost of higher FPR95. Max Entropy is typically intermediate. Overall, MaxLogit provides the best balance between detection power and false positives.

3.2. Confidence calibration

We then studied **Temperature Scaling (TS)** as post hoc calibration. Temperature Scaling is a post-processing approach that rescales the logit layer of a deep model by a parameter T called temperature. TS is used to soften the output of the softmax layer and make it more calibrated [4].

With the code in Figure 5 we divided the logits by the value T (temperature) before applying the softmax. In general, if $T > 1$ the predictions are less confident, otherwise

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	72.20	29.100	62.510	2.711	64.973	1.747	50.763	7.470	41.823	12.426	82.492
MaxLogit	72.20	38.319	59.336	4.626	48.443	3.301	45.494	9.498	40.300	15.581	73.247
Max Entropy	72.20	31.005	62.593	3.051	65.599	2.581	50.368	8.826	41.523	12.678	82.631

Table 1. Anomaly inference performance comparison of different methods on multiple datasets.

the results will be more precise. The temperature directly affects the confidence of the predictions, aiming to achieve better calibration and improve anomaly detection through confidence-based thresholding. Our goal is not higher raw confidence, but better calibration and separation between ID and OoD.

```
if args.method == 'msp':
    softmax_result = F.softmax(result /
        args.temperature, dim=0)
    anomaly_result = 1.0 - torch.max(
        softmax_result, dim=0)[0]
```

Figure 5. Temperature scaling implementation for MSP

To evaluate the influence of temperature scaling we varied the temperature ($T \in \{0.5, 0.75, 1.1\}$), comparing the results with those of the standard case, i.e. when $T = 1$.

Observing the results we chose for each dataset the value of the temperature for which the best performance was obtained.

From the Table 2 we observe that:

- FS L&F and FS Static show an improvement of AuPRC and a reduction of FPR95 when $T = 1.1$ is applied,
- SMIYC RA-21 and RO-21 don’t improve introducing the temperature scaling.
- Road Anomaly shows only a marginal variation.

Although in theory the values of $T > 1$ make prediction less confident, in our research this behavior turns into an improvement for anomaly detection. In fact, reducing the confidence out-of-distribution pixels are more affected than the in-distribution ones. In this way the anomalous pixels undergo a more marked lowering of the maximum softmax probability. The outcome is a more separable confidence distribution between normal and anomalous regions, with a consequent increase in detection metrics (higher AuPRC and lower FPR95). In other words, the fact that the model becomes less “secure” with $T > 1$ is not a limitation, but a practical advantage for MSP-based anomaly segmentation.

3.3. Void classifier

Cityscapes includes a void/unlabeled label in addition to the 19 evaluation classes. Generally void classes contain pixels to ignore or background, and therefore must not be considered during training or during evaluation. In this phase, however, we were tasked to include this class in the possible output and therefore assign the void label to those objects that did not belong to any other class. Proceeding like this, we will perform the anomaly inference by only selecting the output of the Void class.

To evaluate the effectiveness of this approach, we tested three different semantic segmentation networks: ENet, ERFNet, and BiSeNet. By comparing these models, we can assess whether segmentation accuracy correlates with the ability to detect anomalies using the void class. The results of this evaluation are reported in Table 3.

The results show that ERFNet achieves the best segmentation quality with an mIoU of 68.92, but this does not directly translate into the best anomaly detection performance. BiSeNet provides the strongest results in terms of AuPRC, achieving the highest scores on SMIYC RA-21 (32.181) and Road Anomaly (14.176), and remains competitive across other datasets. ERFNet, while not leading in AuPRC, demonstrates more stable FPR95 performance, particularly on SMIYC RO-21 (29.420) and FS Static (36.340). ENet, despite its lower segmentation accuracy (mIoU = 48.76), occasionally produces results close to or better than ERFNet (e.g., Road Anomaly AuPRC = 10.487), but generally lags behind in both AuPRC and FPR95. Overall, these results highlight that higher segmentation accuracy does not guarantee better anomaly detection, and models like BiSeNet can provide superior OoD detection even if they don’t present the highest mIoU.

3.4. Our extension

Until now, we have tried to evaluate anomaly inference techniques using baseline methods (MSP, MaxLogit, Max-Entropy, TemperatureScaling), but this only works in the inference phase, allowing us to calibrate the uncertainty but not to improve the identification of anomalies. For this reason, we thought of extending this work by using new loss functions, which are capable of guiding the learning of the model, thus improving the separation between in-distribution and out-of-distribution data. In particular, we

Method	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP	72.20	29.100	62.510	2.711	64.973	1.747	50.763	7.470	41.823	12.426	82.492
MSP (t = 0.5)	72.20	27.060	62.730	2.419	63.225	1.280	66.737	6.601	43.475	12.187	82.022
MSP (t = 0.75)	72.20	28.156	62.478	2.566	64.052	1.492	51.848	6.990	42.493	12.319	82.284
MSP (t = 1.1)	72.20	29.409	62.589	2.765	65.523	1.859	50.386	7.686	41.586	12.465	82.621
MSP (Best t)	72.20	29.409	62.589	2.765	65.523	1.859	50.386	7.686	41.586	12.465	82.621

Table 2. Comparison of MSP anomaly detection performance on multiple datasets with different temperature t .

Network	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
ENet	48.76	14.384	87.063	1.441	84.229	1.505	45.005	7.584	67.827	10.487	82.174
ERFNet	68.92	17.120	86.057	2.236	29.420	8.676	42.292	11.343	36.340	8.514	94.529
BiSeNet	25.89	32.181	59.541	8.352	48.095	5.287	57.119	5.355	66.506	14.176	85.485

Table 3. Comparison of anomaly detection performance, using void classifier, across multiple networks.

have considered the Cross-Entropy Loss, Focal Loss, Logit Normalization Loss (LogitNorm) and Enhanced Isotropy Maximization Loss (IsoMax+).

The reference loss used for semantic segmentation is the Cross Entropy Loss, defined as

$$\mathcal{L}_{CE}(x, y) = -\log \frac{\exp(f_y(x))}{\sum_{j=1}^C \exp(f_j(x))} \quad (3)$$

which measures the difference between the predicted distribution and the real one.

The Focal Loss introduces a modulating factor to reduce the impact of easily classifiable samples. It can be written as

$$\mathcal{L}_{Focal}(x, y) = -\alpha (1 - \hat{p}_y)^\gamma \log(\hat{p}_y) \quad (4)$$

where $\hat{p}_y$ is the probability for the correct class, α is a class balancer and γ is a focusing parameter.

The Logit Normalization Loss addresses the overconfidence problem normalizing the logit vector during the training. As shown in [8] the logit norm vector tends to increase during the training. By normalizing the logits, the network will focus on the vector direction, improving the separation of ID and OoD data. So the norm vector became a constant

$$\hat{f}(x) = \frac{f(x)}{\|f(x)\|} \quad (5)$$

and the loss is computed as

$$\mathcal{L}_{LogitNorm}(x, y) = -\log \frac{\exp(\hat{f}_y(x)/\tau)}{\sum_{j=1}^C \exp(\hat{f}_j(x)/\tau)} \quad (6)$$

The Enhanced Isotropy Maximization Loss (IsoMax+) is based on the distance between the feature representation

and the class prototypes. It is defined in [3] as

$$\mathcal{L}_{IsoMax+}(x, y) = -\log \frac{\exp(-|d_s| \|\hat{f}_\theta(x) - \hat{p}_y\|)}{\sum_{j=1}^C \exp(-|d_s| \|\hat{f}_\theta(x) - \hat{p}_j\|)} \quad (7)$$

where $\hat{f}_\theta(x) = \frac{f_\theta(x)}{\|f_\theta(x)\|}$ is the normalized feature, $\hat{p}_j = \frac{p_j}{\|p_j\|}$ is the normalized prototype of the class j and d_s is the distance scale, a parameter learning during training.

4. Experiment and Results

4.1. Training

We train on Cityscapes in a 20-class setup, where label 255 is remapped to class 19 (void). Input images are resized to 512×1024 .

Data augmentation:

- ERFNet: random horizontal flip and ± 2 -pixel translation.
- ENet: random horizontal flip.
- BiSeNet: multi-scale resizing ($s \in \{0.75, 1.0, 1.5, 1.75, 2.0\}$), random 512×1024 crops, and ImageNet mean/std normalization.

For the loss function we evaluate several alternatives:

- Cross-Entropy (CE),
- Focal Loss ($\gamma = 2$),
- Logit Normalization (LogitNorm): CE on ℓ_2 -normalized logits, $s = 1$,
- Enhanced Isotropy Maximization (EIM),
- Hybrid variants, such as CE+0.1-EIM or Focal+2-LogitNorm.

Method	Loss	mIoU	SMIYC RA-21		SMIYC RO-21		FS L&F		FS Static		Road Anomaly	
			AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95	AuPRC	FPR95
MSP												
	CE	68.92	21.577	74.044	4.061	19.758	6.922	63.239	8.007	36.987	10.596	87.848
	FOCAL	66.24	24.699	66.436	5.709	29.011	3.595	92.907	7.997	69.513	13.456	69.979
	LN + CE	68.97	21.816	68.725	10.131	26.508	4.804	60.787	12.502	44.979	12.184	74.018
	LN + FL	65.07	19.828	75.958	3.921	26.855	3.254	65.779	6.526	58.049	10.733	76.236
	IsoMax + CE	67.58	24.302	64.810	3.429	53.562	4.024	53.021	10.883	45.997	15.930	76.519
	IsoMax + FL	66.96	23.506	80.575	5.713	80.617	2.785	95.854	10.080	63.326	14.250	73.910
MaxLogit												
	CE	68.92	22.779	70.515	3.455	20.894	2.870	76.238	9.948	42.509	12.769	80.359
	FOCAL	66.24	33.014	53.141	7.774	31.210	1.566	93.352	7.746	62.247	13.525	76.963
	LN + CE	68.97	18.473	80.773	5.304	31.249	5.620	60.592	14.049	44.929	11.868	75.055
	LN + FL	65.07	15.530	96.232	3.010	81.346	3.749	64.242	7.128	59.036	10.324	93.117
	IsoMax + CE	67.58	23.782	63.520	5.600	50.977	5.193	58.204	10.548	49.259	18.952	73.556
	IsoMax + FL	66.96	30.107	68.063	11.387	75.856	1.575	95.291	9.206	70.714	16.824	74.126
Max Entropy												
	CE	68.92	21.456	74.074	3.841	19.754	10.582	62.693	8.895	36.862	10.433	88.065
	FOCAL	66.24	25.323	65.047	7.864	27.491	6.554	93.522	10.193	70.442	14.023	69.334
	LN + CE	68.97	21.289	68.776	8.175	26.507	5.020	60.780	12.839	45.060	12.138	74.067
	LN + FL	65.07	17.312	80.815	3.642	26.853	3.810	65.764	7.169	57.963	10.681	78.813
	IsoMax + CE	67.58	23.620	64.794	3.887	54.201	5.777	51.946	12.581	45.305	16.808	76.735
	IsoMax + FL	66.96	24.435	80.106	6.779	80.587	5.015	96.003	13.951	68.498	14.838	73.837

Table 4. Results obtained training ERFNet using different methods and loss combination

Supervision is class-weighted, following the fixed weights defined per model.

Optimization:

- **ERFNet:** Adam ($\text{lr } 5 \times 10^{-4}$, $\varepsilon = 10^{-6}$, weight decay 10^{-4}), with *epoch-wise* polynomial decay (power 0.9).
- **ENet:** Adam ($\text{lr } 5 \times 10^{-4}$, weight decay 2×10^{-4}), with *iteration-wise* polynomial decay.
- **BiSeNet:** SGD ($\text{lr } 2.5 \times 10^{-2}$, momentum 0.9, weight decay 10^{-4}), with *iteration-wise* polynomial decay.

All models are trained for 100 epochs, using automatic mixed precision (AMP, disabled for the ERFNet encoder) and gradient clipping at 5.0. For the extension, we adopt a two-stage protocol with ERFNet: 100 epochs with the encoder only, followed by 100 epochs with the full encoder-decoder. This procedure is repeated independently for each loss function.

4.2. Results

We decided to analyze the performance on an ERFNet network, using different combinations of the loss functions described before and applying different methods. The results are described in Table 4.

In general, it can be observed that the introduction of new loss as alternative of the cross-entropy allows one to improve the anomaly detection metrics. Segmentation quality remains stable across all configurations (mIoU in the 65-69 interval), with the best score achieved by LN+CE (68.97). The main differences emerge in OOD detection metrics. For SMIYC RA-21, MaxLogit with Focal loss provides the

best trade-off, achieving the highest AuPRC (33.01) and the lowest FPR95 (53.14). In SMIYC RO-21, the highest AuPRC is obtained with MaxLogit using IsoMax+FL (11.39), while the lowest FPR95 is achieved by Max Entropy using CE (19.75). In Fishyscapes L&F, the best AuPRC is given by Max Entropy using CE (10.58), while Max Entropy using IsoMax+CE yields the lowest FPR95 (51.95). For Fishyscapes Static, MaxLogit with LN+CE achieves the best AuPRC (14.05), while Max Entropy using CE obtains the lowest FPR95 (36.86). Finally, in the Road Anomaly benchmark, MaxLogit with IsoMax+CE attains the best AuPRC (18.95), and Max Entropy using Focal provides the lowest FPR95 (69.33). Overall, the results highlight that MaxLogit is the most effective scoring function to maximize AuPRC, while Max Entropy often achieves the lowest FPR95. Among the loss functions, Focal and IsoMax consistently enhance OOD detection performance without significantly compromising segmentation quality. These findings suggest that the selection of the scoring method should be guided by the desired trade-off between recall (AuPRC) and false positive control (FPR95).

5. Conclusions

In this work, we studied real-time anomaly segmentation for street scenes using lightweight architectures suitable for implementation in resource-constrained environments.

Starting from ERFNet as a backbone, we evaluated different anomaly detection strategies, with and without temperature scaling. Our results confirmed that MaxLogit consistently achieves the best trade-off between detection and accuracy, while temperature scaling has proven useful for

improving calibration for certain datasets (FS L&F and FS Static). We explored the use of the void class as a proxy for anomaly detection across multiple lightweight networks, highlighting that high segmentation accuracy is not necessarily related to stronger anomaly detection performance. We extended the study to training-level improvements by analyzing recent loss functions such as Logit normalization and Enhanced Isotropy Maximization. These methods showed promising improvements in OoD detection, keeping segmentation quality stable without increasing model complexity. Overall, our findings suggest that the combination of lightweight backbones, well-chosen scoring functions, and the exploitation of void-class supervision enables efficient and effective segmentation of anomalies in safety-critical applications such as autonomous driving.

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